



Perry Nuclear Power Plant
Rod Penfield
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January 23, 2025
L-24-277

10 CFR 50.73(a)(2)(i)(B)
10 CFR 50.73(a)(2)(v)(D)

ATTN: Document Control Desk
U. S. Nuclear Regulatory Commission
Washington, DC 20555-0001

SUBJECT:
Perry Nuclear Power Plant
Docket No. 50-440, License No. NPF-58
Licensee Event Report Submittal

Enclosed is Licensee Event Report (LER) 2024-005, "Division 3 Diesel Generator Inoperable due to Failure of the Right Bank Air Start Motor." There are no regulatory commitments contained in this submittal.

If there are any questions or if additional information is required, please contact Mr. Robert Oesterle, Manager – Regulatory Compliance, at (419) 321-7462.

Sincerely,

A handwritten signature in black ink, appearing to read "Rod L. Penfield". The signature is stylized with a large "R" and "P".

Rod L. Penfield

Enclosure:
LER 2024-005

cc: NRC Project Manager
NRC Resident Inspector
NRC Region III Regional Administrator

Enclosure
L-24-277

Licensee Event Report 2024-005



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form)

<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocoll.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Perry Nuclear Power Plant

☒ 050☐ 052

2. Docket Number

00440

3. Page

1 OF 3

4. Title:

Division 3 Diesel Generator Inoperable due to Failure of the Right Bank Air Start Motor

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
11	26	2024	2024	- 005 -	00	01	23	2025	Facility Name	☐ 052	Docket Number

9. Operating Mode

1

10. Power Level

100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or in NRC Form 366A).

12. Licensee Contact for this LER

Licensee Contact

George Dujanovic, Staff Nuclear Engineering Specialist, Regulatory Compliance

Phone Number (Include area code)

(440) 280-5200

13. Complete One Line for Each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
A	LC	MSTR	I075	Y					

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On November 26, 2024, at 1158, with the reactor in Mode 1 and 100 percent rated thermal power, the Division 3 diesel generator (DG) was declared inoperable per Technical Specification (TS) 3.8.1, due to the right bank air start motor not operating properly during a monthly surveillance test. Subsequent troubleshooting found that the right bank air start motor air hoses were improperly connected during maintenance in June of 2024. Satisfactory testing of the DG, following proper air hose connection, resulted in correct operation of the right back air start motor, and the Division 3 DG was declared operable at 1731 on November 27, 2024.

The cause was determined to be insufficient use of verification practices and/or use of vendor drawings during the air start motor air hose re-assembly during previous maintenance.

Corrective actions include adding an independent verification procedure step to ensure the proper connection of the air start motor air hoses and revisions to maintenance procedures for configuration control.

This event was reported on November 26, 2024, as a loss of safety function under 10 CFR 50.72(b)(3)(v)(D), for accident mitigation, due to the Division 3 DG being declared Inoperable.



**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME Perry Nuclear Power Plant	☒ 050	2. DOCKET NUMBER 00440	3. LER NUMBER		
	☐ 052		YEAR 2024	SEQUENTIAL NUMBER - 005	REV NO. - 00

NARRATIVE

Energy Industry Identification System (EIIIS) codes are identified in the text as [XX].

BACKGROUND:

The High Pressure Core Spray (HPCS) [BG] system power source, the Division 3 DG [DG], is self-contained, except for access to the preferred source of offsite power through the onsite AC power distribution system and the system actuation signal source. The system is operable as an isolated system independent of electrical connection to any other system by using the Division 3 DG.

HPCS provides and maintains an adequate coolant inventory inside the reactor vessel to limit fuel cladding temperatures in the event of breaks in the reactor coolant pressure boundary. The system is initiated by either high pressure in the drywell or low water level in the reactor vessel [RCT]. It operates independently of all other systems over the entire range of pressure differences from greater than normal operating pressure to zero. The HPCS cooling decreases vessel pressure to enable the low-pressure cooling systems to function. The HPCS system pump [P] motor [MO] is powered by a diesel generator if auxiliary power is not available; the system may also be used as a backup for the Reactor Core Isolation Cooling (RCIC) [BN] system.

DESCRIPTION OF EVENT:

On November 26, 2024, at 1158, with the reactor in Mode 1 and 100 percent rated thermal power, during a surveillance test of the Division 3 DG, it was identified that the right bank air start motor was not operating correctly, based on noting a lack of a sufficient pressure drop at the air storage tank. The Division 3 DG was declared Inoperable, but available, and TS 3.8.1, AC Sources – Operating, Condition B, one required DG inoperable, was entered.

During troubleshooting, it was discovered that the air hoses to the right bank air start motor were not configured correctly, causing improper operation of the right bank air start motor. The air hoses were determined to have been configured incorrectly during maintenance in June of 2024. The air hoses were returned to the proper configuration and a post-maintenance run of the Division 3 DG was performed, resulting in a normal start using both right and left bank air start motors. On November 27, 2024, at 1731, the Division 3 DG was declared Operable and TS 3.8.1 was met.

Although other power sources were available during this period, the inoperable condition of the Division 3 DG is reportable under 10 CFR 50.73(a)(2)(v)(D) for mitigating the consequences of an accident during a loss of offsite power/loss of coolant accident (LOOP/LOCA). Since the air hoses to the right bank air start motor were found to be misconfigured since June of 2024, the Division 3 DG has been inoperable since that time. As such, this is also reportable under 10 CFR 50.73(a)(2)(i)(B).

CAUSE OF EVENT:

Cause:

Insufficient use of verification practices and/or use of vendor drawings to allow for correct re-assembly of the air start motor air hoses.



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NARRATIVE

CORRECTIVE ACTIONS:

Completed Action:

The air hoses to the right bank air start motor were re-connected to the correct configuration.

Scheduled Actions:

- Revise the maintenance procedure step to require independent verification for configuration control purposes.
- Revise the monthly surveillance procedure to record pre-start air bank pressures immediately before starting the DG.
- Revise the monthly surveillance procedure to verify air start motor airflow by checking for exhaust air during the start signal.

EVENT ANALYSIS:

A probabilistic risk assessment (PRA) evaluation was performed for the November 26, 2024, Division 3 DG air start motor incident. The DG remained available, due to the demonstrated ability of the left bank air start motor to start the Division 3 DG reliably, and within surveillance starting time requirements, since the misposition of the air hoses in June of 2024. The risk of this event is therefore considered to be of very low safety significance in accordance with the Regulatory Guidance.

Although the Division 3 DG was inoperable as per the TS Bases with the right bank air start motor not functioning properly, an engineering evaluation concluded Division 3 DG remained able to perform its safety function to achieve proper speed and output voltage within the required start time from June through November 2024. Therefore, in accordance with NEI 99-02, Regulatory Assessment Performance Indicator Guideline, Section 2.2, Mitigating Systems Cornerstone for Safety System Functional Failures, this event will not be counted against the Reactor Oversight Process (ROP) Safety System Functional Failure Performance Indicator.

Site auxiliary power was available during the inoperability period and RCIC remained operable.

PREVIOUS SIMILAR EVENT:

No other similar failures of the Division 3 DG or it's right bank air start motor could be found during the previous three-year timeframe.